

DA-QAIDAM, China

WMO: 527130

Lat: 37.8523N

Lon: 95.3528E

Elev: 10413

StdP: 9.95

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.0	(c) -4.3	(d) -28.1	(e) 1.7	(f) 9.4	(g) -24.3	(h) 2.1	(i) 10.0	(j) 16.8	(k) 20.1	(l) 13.2	(m) 22.5	(n) 1.3	(o) 40	(p) 0.611

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 18.2	(c) 79.1	(d) 51.6	(e) 75.9	(f) 50.0	(g) 73.0	(h) 48.7	(i) 55.3	(j) 70.9	(k) 53.7	(l) 68.5	(m) 52.0	(n) 66.5	(o) 5.8	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 50.1	(b) 79.7	(c) 60.3	(d) 47.8	(e) 73.1	(f) 60.4	(g) 45.4	(h) 66.6	(i) 59.8	(j) 28.2	(k) 70.7	(l) 27.0	(m) 68.3	(n) 25.7	(o) 66.2	(p) 62.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	15.2	12.7	-14.6	84.6	4.7	3.4	-18.0	87.0	-20.7	89.0	-23.3	90.9	-26.7	93.3
			-15.7	57.2	4.4	2.2	-18.8	58.8	-21.3	60.1	-23.8	61.3	-27.0	62.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.3	12.3	19.6	28.5	39.7	49.2	57.1	62.4	61.0	51.5	37.3	24.5	14.7		
	DBStd	18.38	6.34	7.11	6.87	6.76	5.59	4.93	5.49	5.21	5.63	6.13	6.09	5.43		
	HDD50	5382	1169	851	665	312	82	6	0	1	44	395	764	1093		
	HDD65	9824	1634	1271	1130	758	491	239	118	146	406	859	1214	1558		
	CDD50	1098	0	0	0	5	56	221	384	342	89	1	0	0		
	CDD65	65	0	0	0	0	0	4	37	23	1	0	0	0		
	CDH74	501	0	0	0	0	1	27	289	175	9	0	0	0		
	CDH80	70	0	0	0	0	0	0	49	21	0	0	0	0		
(14) Wind	WSAvg	4.4	2.8	3.7	5.1	5.8	6.0	5.5	5.1	4.6	4.3	3.9	3.1	2.7		
(15) Precipitation	PrecAvg	3.4	0.1	0.1	0.1	0.1	0.5	0.8	0.8	0.6	0.3	0.0	0.0	0.0		
	PrecMax	6.6	0.2	0.4	0.7	0.8	2.6	2.3	3.0	2.6	1.3	0.4	0.2	0.2		
	PrecMin	1.3	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.3	0.1	0.1	0.2	0.2	0.6	0.5	0.7	0.6	0.3	0.1	0.0	0.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.4	46.0	57.1	65.9	71.8	77.3	85.4	82.8	74.9	62.7	48.6	37.8		
		MCWB	24.1	28.9	34.7	39.9	43.3	50.4	56.3	54.1	50.3	39.5	31.0	25.1		
	2%	DB	32.2	41.4	50.6	61.5	67.7	73.6	80.3	78.6	70.2	57.5	44.5	33.2		
		MCWB	21.1	26.2	31.2	37.7	41.9	47.5	52.7	51.3	47.0	36.7	28.9	21.9		
	5%	DB	28.3	37.1	46.5	57.8	64.8	70.8	77.0	75.3	66.5	54.0	41.1	29.7		
		MCWB	18.6	24.0	29.4	36.1	40.6	45.8	50.5	50.1	44.6	34.9	27.0	19.8		
	10%	DB	25.0	33.3	42.6	54.3	61.9	68.4	73.8	72.3	63.3	50.8	37.9	26.6		
		MCWB	16.7	21.9	27.6	34.5	39.3	44.8	49.5	48.7	42.9	33.3	25.2	17.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.5	29.1	34.8	41.2	45.5	55.1	58.2	57.1	53.2	41.7	31.7	25.6		
		MCDB	36.8	45.0	55.7	62.3	63.6	69.4	79.5	76.1	69.8	56.7	46.8	37.0		
	2%	WB	21.1	26.5	31.9	38.5	43.6	51.3	55.8	55.0	49.9	38.0	29.2	22.2		
		MCDB	31.5	40.4	48.6	58.9	61.5	65.5	72.1	70.6	63.8	55.1	43.4	32.5		
	5%	WB	18.8	24.3	29.9	36.8	42.2	49.1	54.0	53.6	47.4	35.6	27.2	20.0		
		MCDB	27.6	36.4	45.6	56.1	59.6	63.3	69.4	67.6	62.0	52.8	40.4	29.2		
	10%	WB	16.8	22.1	27.9	35.0	40.8	47.4	52.4	52.0	45.1	33.7	25.4	18.0		
		MCDB	24.7	32.8	42.1	53.0	57.9	61.9	67.6	65.8	59.6	49.9	37.4	26.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	25.7	24.9	24.5	23.6	21.0	18.7	18.2	19.7	21.6	24.6	25.2	25.3		
		MCDBR	28.9	29.4	30.0	28.0	25.3	22.3	22.0	23.0	24.4	27.3	28.4	28.0		
	5% WB	MCWBR	20.3	18.5	16.8	13.8	11.2	8.6	8.1	8.6	10.3	13.6	16.7	19.3		
		MCWBR	27.9	27.8	28.0	25.4	20.5	16.8	17.1	16.7	19.4	25.3	27.2	26.9		
(40) Clear-Sky Solar Irradiance	taub		0.224	0.262	0.314	0.368	0.385	0.387	0.372	0.357	0.324	0.273	0.240	0.217		
	taud		2.433	2.314	2.126	1.979	2.005	2.089	2.219	2.247	2.308	2.368	2.415	2.461		
	Ebn at Noon		314	310	299	288	284	283	286	288	293	302	304	310		
	Edh at Noon		26	34	45	56	56	51	44	42	37	31	26	23		
(44) All-Sky Solar Radiation	RadAvg		950	1252	1712	2107	2290	2204	2161	2052	1767	1483	1076	877		
	RadStd		52	58	84	88	111	153	122	120	95	62	62	40		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.01	+1.21	N/A	N/A	N/A	N/A	-272	-356	+95	N/A
(47) Regional (2 neighbors)		+0.68	N/A	N/A	N/A	N/A	N/A	-229	-302	+97	N/A

Nomenclature: See separate page